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ME 3902 Test 2 Task B: Reading Summary

Reproducibility & Replicability

Reproducibility and replicability are terms that can mean different things depending on the scientific discipline it's being applied to [1]. Scientific research has vastly expanded in the past few centuries, both in breadth and depth, leading to increased utilization of information retrieval, machine learning, and artificial intelligence when trying to sift through research. In addition, mounting pressure to publish research papers in premier journals like Nature and Science can lead some researchers to overstate their findings for greater chance at a tenure position at a university or to apply for grant money that is being spread increasingly thin due to government research budgets not keeping up with growing fields [1]. This is likely to introduce bias, either consciously or unconsciously, as figures and findings can be overinflated to sway perceptions of university and grant committees. Since at least the 1970s, researchers have sought to alleviate this through many different routes, from placing significant value on masking/blinding when recording studies, appropriately large sample sizes, and meta-analysis summaries (utilizing data from multiple studies to study a common phenomenon) [1]. Each of these strategies seek to improve the reproducibility and replicability of their studies. In the 1990s, Jon Claerbout started the "reproducible research movement" wherein he made significant use of the growing computational tools available both to high-level researchers as well as common people. He was one of the pioneers of open-source coding, believing that due to the significant weight a program's code carries, it would be beneficial to have it publicly available alongside the study's results, allowing it to be peer-reviewed and getting a better look at how its conclusions were drawn [1].

As mentioned earlier [1], reproducibility and replicability have different definitions based on what scientific discipline/institution it's being referred to in. In fields like Political Science, "replication" has become the standard and encompasses all terms' meanings [1]. In other settings, it may require a qualifying term to narrow down its meaning, with terms like "methods reproducibility", "results reproducibility", and "inferential reproducibility" [1]. The computer science industry has had difficulty classifying these, struggling to compare its industry-specific usage to that in related industries. The specific issue was how "reproducible research" was coined as any research that – if given the data and code – could be reproduced by another researcher, differing from a definition that states a new researcher gathers new data in an attempt to independently verify a previous study [1]. There remains 3 main categories in which the terms are used:

A – They are used with no distinction between them.

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B1 – “Reproducibility” is an instance where the original researcher’s data and code is used to regenerate results, and “replicability” is used to refer to when a researcher’s new data supports findings from a previous study.

B2 – “Reproducibility” means multiple researchers came to the same conclusion using their own data/methods, while “replicability” refers to a different team coming to the same conclusion with the same original methods/code.

This leads to a lot of confusion as these terms must be defined prior to the research being presented.

Gage Repeatability & Reproducibility (GR&R) in Mechanical Testing

While the National Academies report focuses on how the terms are defined across disciplines, the Instron white paper shows how repeatability and reproducibility are given precise, statistical meanings in the world of mechanical testing [2]. Gage Repeatability and Reproducibility (GR&R) is a type of statistical analysis performed by quality and product engineers to validate and verify test equipment, and such studies are recommended by quality plans like the Automotive Industry Action Group (AIAG), Six Sigma, and ISO 9000 [2]. In this context, “repeatability” defines how well a single system can produce a known result over multiple tests, while “reproducibility” is the ability of a different operator to produce the same results from similar parts with the same level of consistency [2]. The purpose of the study is to quantify how much of the observed variability comes from the test system itself versus actual part-to-part variation or process changes. The output is a single quantitative value that falls into one of three categories: a GR&R value under 10% is ideal and suggests system variability is negligible, a value between 10% and 30% is not negligible but may still be acceptable for evaluating part variability, and a value above 30% means the error in the system is too great to differentiate system error from part variation [2]. It is worth noting these ranges were originally developed for non-destructive testing in the automotive industry, and there is some question as to whether they are appropriate for destructive materials testing, where the statistical analysis becomes significantly more complex [2].

An important warning from the paper is that GR&R alone does not address accuracy – it is entirely possible for a test system to produce very consistent (low GR&R) results that are simply wrong [2]. Because of this, the authors argue a test system must be examined both quantitatively, through the GR&R study itself, and qualitatively, using a fishbone diagram of common error sources organized into six major categories: method, measurement, operator, material, machine, and environment [2]. To demonstrate this in practice, the authors conducted a non-destructive GR&R study on four electromechanical test systems from four different manufacturers, using three springs of varying stiffness compressed to a fixed displacement while two operators each tested each spring ten times [2]. Even in this deliberately simplified compression test, the mean peak loads for each spring varied by roughly 24 to 26 pounds between systems. The systems with GR&R values under 10% owed their performance to well-manufactured frames and control electronics and an automatic pre-load feature that removed manual steps from the operator [2]. The poorer-performing systems revealed a variety of error sources: a low data acquisition rate (5 points per second versus 100 on the best system, meaning the software had to interpolate between sparse data points), unverified speed accuracy, control electronics that overshot the target pre-load, and low system stiffness (compliance), which caused one system to consistently report lower peak loads than the others [2]. Notably, the authors found that when total system error is low, the operator is typically the greatest remaining source of error, which is why detailed operating procedures, regular training, and test methods that minimize manual steps are so important [2]. Their overall recommendation is to resolve the qualitative error sources first, and to confirm a system can pass a non-destructive GR&R under 10% before attempting a destructive study, since a system that fails the simplified test will never produce acceptable destructive results [2].

Conclusion

Taken together, the two readings show the same underlying problem at two very different scales. The National Academies report demonstrates that even the definitions of reproducibility and replicability are unsettled across scientific disciplines, which complicates any attempt to evaluate whether research holds up [1]. The Instron paper shows how the mechanical testing industry has responded to this ambiguity by assigning the terms strict statistical definitions and pass/fail thresholds through GR&R studies [2]. In both cases, the lesson is the same: results cannot be trusted, compared, or built upon until the sources of variation – whether they come from researcher bias and unpublished code, or from data rates, compliance, and operator technique – are identified, defined, and controlled.

Works Cited

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[2] M. Viverios and J. Richey, "GR&R: Understanding Sources of Error in Mechanical Testing Results," Instron, 25 September 2007. [Online]. Available: https://www.instron.com/wp-content/uploads/2024/07/gage-rr-articlewhitepaper.pdf. [Accessed 9 June 2026].